Agentic and Communal Wording in Job Advertisements: A Lexicon-Based Analysis Across Male- and Female-Dominated Occupations

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Abstract

Prior work argues that subtly gendered wording in job advertisements can help sustain occupational gender segregation. We test whether the balance of *agentic* (masculine-coded) and *communal* (feminine-coded) language differs between male-dominated and female-dominated occupations in a large, public corpus of real online job ads. Using 17,014 legitimate postings from the Employment Scam Aegean Dataset, we scored each advertisement with the gendered word-stem lists of Gaucher, Friesen, and Kay and measured the rate of agentic and communal words per 100 words. Ads in male-dominated functions (engineering, information technology) used significantly more agentic language than ads in female-dominated functions (administrative, health-care, human resources, education), $t(2834) = 7.63$, $p < .001$, Cohen's $d = 0.24$. Communal wording, however, did not differ between the two groups, $p = .465$. A one-way ANOVA across six functions confirmed that net agentic wording varies by occupation, $F(5, 4291) = 11.29$, $p < .001$, partial $\eta^2 = .013$. The gender gap in wording therefore operates primarily through agentic rather than communal language, with small effect sizes consistent with the subtlety of the phenomenon.

1 Introduction

Women remain underrepresented in many male-dominated fields such as engineering and information technology. Beyond overt barriers, the *language* of recruitment materials may play a role. Gaucher, Friesen, and Kay [1] proposed that job advertisements contain systematically gendered wording: *agentic* or masculine-coded words such as *competitive*, *dominant*, and *assertive*, and *communal* or feminine-coded words such as *supportive*, *collaborative*, and *warm*. In their experiments, masculine-coded wording reduced how appealing a job seemed to women, chiefly by lowering their sense of belonging. Because such wording appears more often in male-dominated occupations, it may act as a subtle, institutional-level mechanism that helps maintain gender inequality.

Most evidence for this claim comes from experiments or from limited archival samples, and its generality is not settled. A recent replication by Seong and Parker [2] reproduced the original effect for start-up job adverts but not for adverts from established firms, suggesting that the phenomenon may be more context-dependent than initially assumed. We therefore revisit it at scale using a large corpus of real online job advertisements, asking whether the balance of agentic and communal wording tracks the gender composition of the occupation. We examine three hypotheses, stated here in prose. First, advertisements in male-dominated occupations will contain a higher rate of agentic words than advertisements in female-dominated occupations. Second, advertisements in female-dominated occupations will contain a higher rate of communal words than advertisements in male-dominated occupations. Third, the net agentic wording of an advertisement (agentic minus communal) will differ systematically across occupational functions rather than being constant.

2 Method

2.1 Data source and sample

We used the Employment Scam Aegean Dataset (EMSCAD), a publicly available corpus of 17,880 real-life English-language job advertisements published through a recruitment platform and released by Vidros et al. [3]; we obtained it from its public Kaggle distribution [4], which is released under a CC0 (public domain) licence. The dataset labels each ad as legitimate or fraudulent and includes a free-text *description*, a *requirements* field, and a categorical *function* (occupation) label. We discarded the 866 fraudulent ads and kept only legitimate postings that carried a function label and a non-empty description, and we removed very short ads (fewer than 30 word tokens). The analytic sample for the primary comparison contained 4,297 advertisements across six focal functions.

2.2 Measuring gendered wording

We operationalized gendered wording with the masculine- and feminine-coded word stems published by Gaucher et al. [1], using the machine-readable version distributed in the open-source *gender-decoder* tool [5] (52 agentic stems, 50 communal stems). Following that tool's matching rule, each advertisement was lowercased and tokenized, and a token was counted as agentic (or communal) if it began with any agentic (or communal) stem; this stem-prefix rule captures morphological variants such as *compete*, *competitive*, and *competitor*. For each ad we computed the number of agentic and communal words per 100 word tokens, which normalizes for advertisement length, together with a *net agentic* score defined as the agentic rate minus the communal rate.

2.3 Grouping occupations by workforce gender

We classified functions by their workforce gender composition, using U.S. Bureau of Labor Statistics data from the Current Population Survey [6]. Under these figures, engineering (about 17% women) and computer and information technology (about 26% women) are male-dominated, whereas health-care practitioner and technical roles (about 76% women), education (about 73% women), and administrative support are female-dominated. We therefore treated engineering and information technology as male-dominated, and administrative support, health-care provision, human resources,

Table 1: Agentic and communal words per 100 words by function. SD in parentheses; *Net* is agentic minus communal.

Function	n	Agentic	Communal	Net
Male-dominated				
Engineering	1227	0.93 (0.80)	1.01 (0.78)	−0.08
Information Technology	1706	1.01 (0.97)	1.18 (1.02)	−0.17
Female-dominated				
Health Care Provider	336	1.00 (0.79)	1.16 (0.86)	−0.16
Human Resources	194	1.08 (0.84)	1.07 (0.83)	0.02
Education	324	0.43 (0.92)	0.83 (1.41)	−0.40
Administrative	510	0.70 (0.74)	1.21 (1.10)	−0.51

and education as female-dominated.[1] These six functions were retained both for the two-group comparison and for a finer-grained analysis across individual functions.

2.4 Statistical analysis

For each dependent measure we compared the male-dominated and female-dominated groups with independent-samples t tests. Because Levene's test indicated unequal variances for the communal measure, we report Welch's t throughout, which does not assume equal variances, and we report Cohen's d as the effect size. As a distribution-free robustness check we also ran Mann–Whitney U tests. To test the third hypothesis we ran a one-way ANOVA of the net agentic score across the six functions, reporting the F statistic and partial η^2. Analyses used SciPy [7].

3 Results

3.1 Descriptive statistics

Table 1 reports agentic and communal wording rates for each of the six functions. Communal words were somewhat more frequent than agentic words in most functions, but the balance shifted toward agentic language in the male-dominated functions.

3.2 Agentic wording (Hypothesis 1)

A Welch independent-samples t test supported the first hypothesis. Advertisements in male-dominated functions ($M = 0.98$, $SD = 0.91$, $n = 2933$) contained significantly more agentic words per 100 words than advertisements in female-dominated functions ($M = 0.76$, $SD = 0.85$, $n = 1364$), $t(2834) = 7.63$, $p < .001$, Cohen's $d = 0.24$. The nonparametric Mann–Whitney test agreed ($p < .001$). Figure 1 shows the group means.

3.3 Communal wording (Hypothesis 2)

The second hypothesis was not supported. Communal wording did not differ significantly between male-dominated ($M = 1.11$, $SD = 0.93$) and female-dominated ($M = 1.09$, $SD = 1.11$) advertisements, $t(2290) = 0.73$, $p = .465$, Cohen's $d = 0.03$. Female-dominated occupations were therefore not characterized by more communal language than male-dominated ones.

[1]This classification is an external benchmark: it labels occupations by their national workforce composition, not by the gender of any applicant, which the corpus does not record. The EMSCAD advertisements date from 2012–2014 and are not exclusively from the United States, but the gender composition of these fields has been highly stable over time, so the benchmark remains applicable.

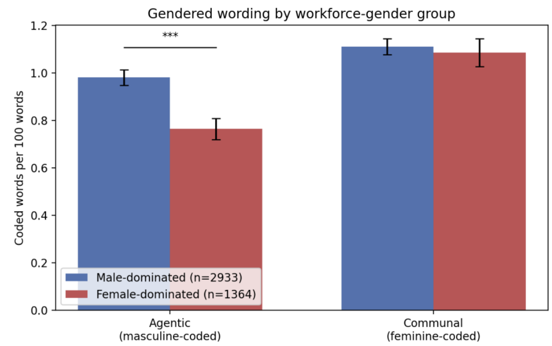

Figure 1: Agentic and communal words per 100 words by workforce-gender group. Error bars are 95% confidence intervals. The agentic difference is significant ($^{***}p < .001$); the communal difference is not.

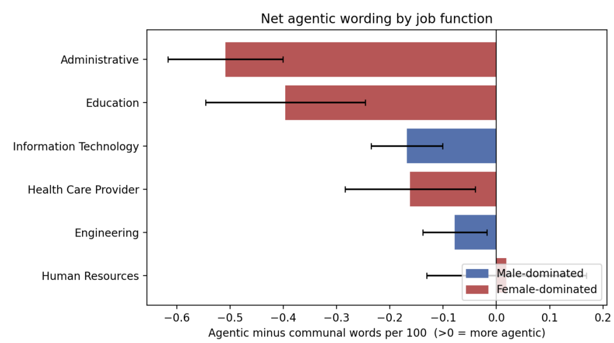

Figure 2: Net agentic wording (agentic minus communal words per 100 words) by function. Positive values indicate more agentic than communal wording. Error bars are 95% confidence intervals.

3.4 Variation across functions (Hypothesis 3)

Consistent with the third hypothesis, a one-way ANOVA showed that the net agentic score differed across the six functions, $F(5, 4291) = 11.29$, $p < .001$, partial $\eta^2 = .013$. As Figure 2 shows, engineering and information technology leaned most toward agentic language, whereas administrative and education roles leaned most toward communal language, although the overall effect size was small.

4 Discussion

Across more than four thousand real job advertisements, the balance of gendered wording tracked the gender composition of the occupation, but only partially. Male-dominated technical fields used measurably more agentic language than female-dominated fields, echoing the pattern that Gaucher et al. [1] argued can discourage women from applying. Contrary to expectation, communal wording did not distinguish the groups: female-dominated occupations were not written in more communal terms. The gendered signal in these ads is thus asymmetric, carried by the presence of agentic language rather than by the absence or presence of communal language.

The effect sizes were small ($d = 0.24$ for the agentic comparison; partial $\eta^2 = .013$ across functions). This is consistent with the original framing of gendered wording as a *subtle* cue rather than a blatant one, and it cautions against overinterpreting the difference. Even small differences, however, may accumulate across the many advertisements a job seeker encounters. Our partial and asymmetric findings also align with the emerging picture that the gendered-wording effect is context-dependent rather than uniform, as in Seong and Parker's [2] replication, which held for start-ups but not for established firms.

Several limitations apply. The lexicon is a fixed English word list and counts words out of context, so it cannot capture negation, phrasing, or domain- specific meaning (for example, *analyze* is coded as agentic regardless of how it is used). The occupation groups are coarse, and the classification into male- and female-dominated relies on external labor statistics rather than on the gender of any applicant. The corpus reflects one recruitment platform and the 2012–2014 period, so results may not generalize to other markets or to the present. Finally, the design is observational and correlational: it describes an association between occupation and wording, not a causal effect of wording on who applies.

5 Conclusion

Using a large public corpus of real job advertisements, we found that male-dominated technical occupations use more agentic language than female-dominated occupations, while communal language does not differ between them. The gendered wording documented in earlier experimental work is detectable at scale, but it is small and asymmetric, concentrated in agentic rather than communal vocabulary.

References

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